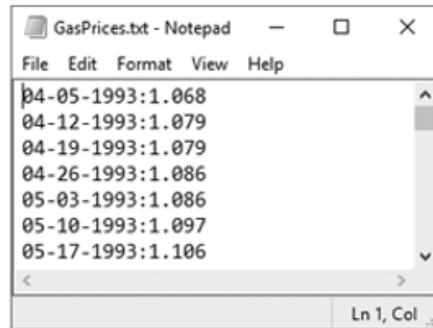


CSE 112 – Gas Prices – STL Containers and String Tokenizing

Design and develop a program that reads the file GasPrices.txt (attached) into an STL container or containers and computes the values and displays the output as shown below. The file contains weekly average gas prices for a gallon of gas in the U.S., beginning on April 5, 1993, and ending August 26, 2013 (a few lines from the file shown here).



Each line of the file is formatted MM-dd-yyyy:price (see chapter 10 for string tokenizing using `split()` or use `std::string delim = ":"`; and `string.find(delim)`). To convert a string to a double, use `stod(string)`.

Design and develop the program to read the contents of the file and extract the data into one or more STL containers appropriate for your algorithm. For each year, the program will determine the highest, lowest, and average price per gallon, and display the data and write it to a file named `your-first-initial_lastName.txt` in the same format displayed. Submit the code and output file for grading.

Partial output shown for required output formatting and testing purposes.

```
Year: 1993 - Gas Prices per Gallon
Highest price: $ 1.11
Lowest price: $ 1.00
Average price: $ 1.07

Year: 1994 - Gas Prices per Gallon
Highest price: $ 1.17
Lowest price: $ 0.99
Average price: $ 1.08

Year: 1995 - Gas Prices per Gallon
Highest price: $ 1.25
Lowest price: $ 1.11
Average price: $ 1.16

Year: 1996 - Gas Prices per Gallon
Highest price: $ 1.33
Lowest price: $ 1.13
Average price: $ 1.24
```